

SHREYAS ACHARYA

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EXPERIENCE

PRG Lab UMD

Jan 2024 – June 2024

Graduate Research Assistant

College Park, MD

- Developed an end-to-end reinforcement learning pipeline for 3D object reconstruction using tactile sensing, achieving upto 95% surface coverage (IoU) on unseen objects, demonstrating strong generalization capabilities.
- Engineered a sophisticated reward structure combining area maximization with exploration bonuses, averaging 85%+ coverage on diverse objects with complex geometries, showcasing the system's robustness in real-world applications.
- Implemented a real-time 3D reconstruction pipeline using PyBullet and TACTO simulation environments, reducing exploration time by 40% through optimized action space and early termination conditions, enabling efficient tactile exploration in visually challenging environments.

techR Business Solutions

June 2022 – July 2023

Software Engineer

Pune, IN

- Architected and deployed a real-time Human Activity Recognition system using 3D ResNet-34, achieving over 85% accuracy with 30 FPS processing speed through optimized ONNX model inference.
- Engineered a hybrid multi-threaded application architecture combining desktop GUI with Flask web interface and WebSocket communication, reducing system latency by 40% and enabling cross-platform monitoring.
- Developed an intelligent detection pipeline with context-aware alert mechanisms, incorporating temporal analysis and smart alert triggering, improving detection accuracy by 20% and enabling live monitoring for fall detection systems.

PROJECTS

Temporal Coherence Evaluation for Multimodal Foundation Models

Nov 2024

Python, PyTorch, Hugging Face, CLIP

- Developed a robust evaluation framework for vision-language-action models, assessing video understanding and video captioning tasks using semantic similarity (BERTScore) and a novel reference-free metric (CLIPGain).
- Demonstrated significant gains in temporal understanding with increased frame context; improved semantic alignment by 6.2% (BERTScore) and numerical prediction accuracy by 55% across 200+ video samples.
- Benchmarked six multimodal foundation models, revealing up to 14× improvement in cross-frame consistency (CLIPGain) for models leveraging full video context.

Multiview Structure from Motion for 3D Scene Reconstruction

May 2024

Python, OpenCV, NumPy, SciPy, Open3D

- Designed a 3D reconstruction pipeline to accurately reconstruct scenes from multiple images, employing SIFT/FLANN-based matching for enhanced accuracy across synthetic and real-world datasets.
- Incorporated custom Bundle Adjustment optimization to jointly refine 3D points, reducing reprojection errors by 10x across all datasets while preserving structural integrity, yielding high-fidelity 3D structures.

SKILLS

Languages: Python, C++ , MATLAB, SQL, Javascript

Frameworks & Libraries: TensorFlow, PyTorch, OpenCV, Scikit-Learn, NumPy, Pandas, Open3D, Stable-Baselines3, ROS2, React, NodeJs

Platforms & Tools: Linux, Git, GitHub Actions, Docker, Hugging Face, RESTful APIs, CI/CD, SolidWorks, Cursor dev, AWS, Microsoft Azure, CUDA, Apache Spark, Kafka, Vector Databases

Software Concepts: UML, Software Design, Agile Development, GPU Programming, SLDC

EDUCATION

University of Maryland, College Park

Aug 2023 – May 2025

MEng in Robotics

College Park, MD

Coursework: Multimodal Foundation Models, Computer Processing of Pictorial Information, Robot Learning, Data Structures and Algorithms, Perception of Autonomous Robots

Savitribai Phule Pune University

Aug 2019 – May 2023

B.E in Computer Engineering

Pune, IN